



The 7th Annual

# Parallel Applications Workshop Alternatives to MPI+X

Sunday, November 17, 2024  
9:00 am - 5:30 pm (EST)

Hybrid Event

Held in conjunction with SC24  
The International Conference for High Performance Computing, Networking, Storage, and Analysis





## Parallel Applications Workshop Alternatives to MPI+X

November 17<sup>th</sup>, 2024  
Sunday



# Organization

- Workshop Chair
  - Karla V. Morris Wright - Sandia National Laboratories
- Organizing Committee
  - Engin Kayraklioglu - Hewlett Packard Enterprise
  - Kenjiro Taura - University of Tokyo
  - Irene Moulitsas - Cranfield University
  - Elliott Slaughter - SLAC National Accelerator Laboratory
- Program Committee Chairs
  - Bill Long - Hewlett Packard Enterprise
  - Daniele Lezzi - Barcelona Supercomputing Center
- Advisory Committee
  - Bradford L. Chamberlain - Hewlett Packard Enterprise
  - Damian W. I. Rouson - Lawrence Berkeley National Laboratory



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# Program Committee

- **Chair:** Daniele Lezzi - Barcelona Supercomputing Center
- **Chair:** Bill Long - Hewlett Packard Enterprise
- Marjan Asgari - National Resources Canada
- Scott Baden - University of California, San Diego
- Dan Bonachea - Lawrence Berkeley National Laboratory
- Jan Ciesko - Sandia National Laboratory
- Nelson Dias - Federal University of Paraná
- Mario Di Renzo - University of Salento
- David Eberius - Intel
- Engin Kayraklioglu - Hewlett Packard Enterprise
- Francesc Lordan - Barcelona Supercomputing Center
- Henry Monge Camacho - Oak Ridge National Laboratory
- Karla V. Morris Wright - Sandia National Laboratories
- Irene Moulitsas - Cranfield University
- Catherine Olschanowsky - Advanced Micro Devices, Inc
- Tom Quinn - University of Washington
- Michel Schanen - Argonne National Laboratory
- Michael Schlottke-Lakemper - University of Augsburg
- Elliott Slaughter - SLAC National Accelerator Laboratory
- Kenjiro Taura - University of Tokyo
- Thiago Teixeira - Intel
- Jana Thayer - SLAC National Accelerator Laboratory
- Miwako Tsuji - RIKEN



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# Artifact Evaluation Committee

- **Chair:** Irene Moulitsas - Cranfield University
- **Chair:** Elliott Slaughter - SLAC National Accelerator Laboratory
- Oliver Alvarado Rodriguez - New Jersey Institute of Technology
- Desmond Bisandu - Cranfield University
- Yakup Budanaz - ETH Zurich
- Fabio Durastante - University of Pisa
- Guillaume Helbecque - University of Lille
- Mert Hidayetoglu - Stanford University
- Boyu Neil Kuang - Cranfield University
- Seema Mirchandaney - Stanford University
- Soren Rasmussen - National Center for Atmospheric Research
- Kate Rasmussen - Lawrence Berkeley National Laboratory
- Anjiang Wei - Stanford University



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# Program

- 9:00 - Welcome and Introduction
- 9:02 - **Session 1**
- 10:00 - Break (30 minutes)
- 10:30 - **Session 2**
- 12:30 - Lunch (90 minutes)
- 2:00 - **Session 3: Distinguished Talk**  
Éric Laurendeau - Polytechnique Montréal  
*A Case Study for Using Chapel  
within the Global Aerospace Industry*
- 3:30 - Break (30 minutes)
- 3:30 - **Session 4**
- 3:45 - **Panel Discussion**
- 5:30 - PAW-ATM Concludes



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# Session 1

9:00 - 10:00

- **9:02: Survey of Technologies for Developers of Parallel Applications**

Wonchan Lee – NVIDIA

Damian Rouson – Lawrence Berkeley National Laboratory

- **9:45: User Experience: *Resource Adaptivity at Task-Level***

Jonas Posner



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# Session 2

10:30 - 12:30

- **10:30: *Speeding-Up LULESH on HPX: Useful Tricks and Lessons Learned using a Many-Task-Based Approach***  
Torben Kalkhof, and Andreas Koch
- **10:50: *Lamellar: A Rust-based Asynchronous Tasking and PGAS Runtime for High Performance Computing***  
Ryan D. Friese, Roberto Gioiosa, Joseph Cottam, Erdal Mutlu, Gregory Henselman-Petrusek, Polykarpos Thomadakis, Mark Raugas
- **11:10: *Applying a Task-Based Approach to Distributed Machine Learning Workflows***  
Fernando Vazquez-Novoa, Daniele Lezzi, Francesc Lordan, Fatemeh Baghdadi, and Davide Cirillo
- **11:30: *Accelerating Multi-GPU Embedding Retrieval with PGAS style Communication for Deep Learning Recommendation Systems***  
Yuxin Chen, Aydin Buluc, Katherine Yelick, and John Owens
- **11:50: *Mitigating Synchronization Bottlenecks in High-Performance Actor-Model-Based Software***  
Kyle Klenk, Mohammad Mahdi Moayeri, Junwei Guo, Martyn P. Clark, and Raymond J. Spiteri
- **12:10: *Intel SHMEM: GPU-initiated OpenSHMEM using SYCL***  
Alex Brooks, Philip Marshall, David Ozog, Md. Wasi-ur- Rahman, Lawrence Stewart, and Rithwik Tom



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# Session 3

2:00 - 3:00

## 2:00: Distinguished Talk

*A Case Study for Using Chapel  
within the Global Aerospace Industry*

Éric Laurendeau, Polytechnique Montréal

- **2:45: User Experience: Exploring Suffix Array Algorithms in Chapel**

Michael P. Ferguson, Bonnie Hurwitz, and Shreyas Khandekar





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# Session 4

3:30 - 5:30

- **3:30: User Experience: *Just Write Fortran: Experiences with a Language-Based Alternative to MPI+X***

Baboucarr Dibba, Katherine Rasmussen, Brad Richardson, Damian Rouson, David Torres, Yunhao Zhang, Ethan Gutmann, Kareem Ergawy and Michael Klemm

- **3:45: Panel: Alternative programming models for applications at scale**

Panel Chair: Christine Sweeney – Los Alamos National Laboratory

- Jan Ciesko – Sandia National Laboratories
- Nils Deppe – Cornell University
- Jason DeVinney – Center for Computing Sciences
- Eric Laurendeau – Polytechnique Montreal
- Julian Samaroo – Massachusetts Institute of Technology

# PAW-ATM Journey

